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Note on the Influence of Nitrogen Content on the Resistance to Pitting Corrosion of Stainless Steels

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Pitting potentials have been measured and some gravimetric testing has been carried out on a series of experimental austenitic stainless steels with varying chromium, molybdenum and nitrogen contents. All three of these elements were found to contribute to the resistance to the initiation of pitting corrosion, and synergistic effects have been noted. The effect of nitrogen is especially potent in a steel with 22% chromium and 3% molybdenum.

Introduction

NITROGEN is always present to some extent in stainless steels, usually in small amount as an impurity but sometimes in more significant amount as an alloying addition. In this latter role it has been used as a more economical partial or complete replacement for nickel to promote the presence of austenite, as a solid solution lattice strengthener, or to give increased rate of work hardening with austenitic structures. The effect of nitrogen content on corrosion resistance has, on the whole, been judged as a result of the testing of steels developed with aspects other than corrosion in mind and it is not surprising that conclusions have varied in view of the other changes in composition usually coincident with alloying with nitrogen.

Bibliographies regarding the effect of nitrogen content on corrosion resistance were included in papers by Steenslund[1] and Hartline[2] and, in a proportion of the work reviewed, there was evidence of enhanced resistance to the initiation of pitting corrosion from alloying with nitrogen. In some instances, improvement could have been related to changes in ratios of austenite and ferrite in the structures, but in experiments by Steenslund,[1] Hartline,[2] and Janik-Czachor *et al.*,[3] improvements within single phase (austenitic) alloys were shown.

During alloy development work in the authors' laboratory, involving mainly austenitic steels with more than 20% chromium, more than 1·5% molybdenum and more than 0·15% nitrogen, a significant effect on the resistance to initiation of pitting corrosion from variations in nitrogen content was noted. Moreover it appeared that, when the contents of chromium and molybdenum are sufficiently high, a large increase in pitting resistance can result from a relatively small increase in nitrogen content. Similar 'transitions' to a high resistance were also noted from increase in either chromium or molybdenum with suitable contents of the other significant elements. The range of nitrogen contents covered in the work was rather limited since, with the base composition used, low nitrogen contents would have led to duplex structures. The objective of the work reported here was to investigate the influence of relatively wide variations in the contents of chromium, molybdenum and, especially, of nitrogen, utilising a base composition such that all the alloys were austenitic.

Testing

The basic composition chosen was 0·03% carbon, 0·2% silicon, 4% manganese (to ensure freedom from outgassing during solidification in all cases) and 20% nickel (to ensure a fully austenitic structure in all cases). 5 kg melts were made at atmospheric pressure in an induction furnace using Japanese iron, nickel pellets, electrolytic chromium, ferro-molybdenum, refined manganese, and high-nitrogen ferro-chromium. Ingots were cast and hot rolled to 20 mm diameter bars which were heat treated at 1100°C for 30 min and cooled in still air. Detailed compositions are given in Table I.

A potentiodynamic technique was used to measure the pitting potential in an aerated solution of 0·6 M sodium chloride and 0·1 M sodium bicarbonate. The specimen surface was ground to a 120 grit finish and 5 cm^2 was exposed to the corrodent. The potential was controlled, relative to a standard saturated calomel electrode, by a potentiostat. Following pretreatment for 5 min at $-1·5$ V, the potential was increased rapidly to $-0·5$ V and thereafter at a steady rate of 1 V in 50 min. In this work the critical pitting potential was taken as that at which there was an anodic current flow of 10^{-5} A/cm^2. The values obtained with each steel are given in Table II.

Several workers have reported that, with testing of the type described above, corrosion was wholly or partly at the junction between the exposed surface and the coating used to protect the non-exposed surface (i.e., was a form of crevice corrosion). Such an effect has been found intermittently with testing in the authors' laboratory and, while it appears to have little effect on the values of the pitting potential with samples of the less resistant materials (i.e., with pitting potentials up to about + 150 mV), its occurrence can lead to low results (and 'scatter') with more resistant materials. Since the objective of testing is to determine relative resistances to pitting initiation, one method of restricting scatter due to intermittent crevice corrosion is to ensure the presence of a standard crevice with all samples which should also render the test more aggressive. Thus 'pitting' potentials were also measured for samples having a ring made from round section 'neoprene' (2·5 mm section) fastened firmly round the cylindrical specimen. The results are included in Table II.

Immersion tests without polarisation (duration 1 week) were carried out at ambient temperature in a variety of solutions using samples with a 120 grit finish, 12·5 mm dia. × 32 mm. In all cases any corrosion was of the pitting and/or crevice type, the latter being at points of contact with the test beakers. The corrodents were solutions of ferric chloride (60, 6 or 1 g/l $FeCl_3$) and sodium chloride (200 g/l), the latter having a pH of 2 achieved by adding hydrochloric acid. Losses in weight were determined and are included in Table II.

Discussion

It has long been appreciated that high contents of chromium and of molybdenum lead to improved resistance to initiation of pitting corrosion and this has been further demonstrated recently by use of modern techniques.[4-7] Such beneficial and complementary effects are obvious for the relatively low nitrogen steels in the present series from the comparison of the pitting potentials for steels 1, 3, 7, 11, 13 and 16. The values are somewhat lower than for commercial steels of near-equivalent chromium and molybdenum contents and this is attributable to an adverse effect of manganese,[8] 4% of which was present in all the experimental steels.

TABLE I

Composition of stainless steels, wt-%

Steel	C	Si	Mn	S	P	Cr	Ni	Mo	N
1	0·030	0·13	3·84	0·011	0·014	17·38	20·15	0·02	0·036
2	0·025	0·19	4·22	0·010	0·017	17·68	20·43	0·08	0·300
3	0·035	0·19	4·38	0·010	0·016	17·95	19·90	1·03	0·031
4	0·007	0·22	4·08	0·016	0·020	16·42	19·90	1·03	0·155
5	0·024	0·33	4·06	0·016	0·020	16·86	19·98	1·03	0·270
6	0·025	0·17	4·31	0·012	0·018	16·66	20·54	1·01	0·290
7	0·021	0·18	4·25	0·014	0·019	16·34	19·16	2·98	0·034
8	0·008	0·14	4·06	0·019	0·017	16·70	19·65	2·92	0·149
9	0·012	0·27	4·07	0·018	0·020	16·80	20·05	2·94	0·291
10	0·025	0·23	4·53	0·010	0·019	16·58	20·29	2·72	0·310
11	0·012	0·13	4·28	0·009	0·007	22·58	19·88	0·05	0·054
12	0·017	0·14	4·33	0·008	0·008	22·46	20·27	0·10	0·390
13	0·020	0·16	4·20	0·007	0·015	22·13	20·30	0·99	0·054
14	0·008	0·15	4·02	0·018	0·016	22·52	19·85	1·07	0·155
15	0·011	0·20	3·80	0·015	0·020	22·72	19·95	1·02	0·412
16	0·025	0·21	4·29	0·011	0·013	22·41	19·96	2·64	0·067
17	0·012	0·14	3·80	0·019	0·017	22·84	20·05	2·85	0·159
18	0·022	0·16	4·35	0·011	0·015	22·35	20·39	2·73	0·310
19	0·020	0·23	3·75	0·017	0·019	22·52	19·95	2·96	0·455

TABLE II

Breakdown potentials and weight losses for stainless steels in various solutions

Steel	Composition variants			Breakdown potentials in aerated NaCl + $NaHCO_3$, mV (SCE)		Weight losses in tests (unpolarised) in various solutions, mg/cm^2/week			
	% Cr	% Mo	% N	Plain	Crevice	$FeCl_3$ 60 g/l	$FeCl_3$ 6 g/l	$FeCl_3$ 1 g/l	NaCl 200 g/l (pH 2)
1	17·38	0·02	0·036	− 65	− 90	82·9	6·3	1·30	0·7
2	17·68	0·08	0·300	− 25	− 15	81·9	6·9	1·40	0·5
3	17·95	1·03	0·031	− 40	− 30	81·6	6·1	0·90	0·3
4	16·42	1·03	0·155	− 15	− 60	83·1	6·7	1·30	0·5
5	16·86	1·03	0·270	+ 65	− 20	85·8	6·6	1·25	0·4
6	16·66	1·01	0·290	+ 50	+ 20	80·3	6·5	1·25	0·2
7	16·34	2·98	0·034	+ 20	+ 10	83·7	3·4	0·50	0·2
8	16·70	2·92	0·149	+ 85	+ 40	80·4	3·6	0·00	0·3
9	16·80	2·94	0·291	+ 80	+ 95	57·7	0·3	0·50	0·1
10	16·58	2·72	0·310	+ 95	+ 70	86·7	0·7	0·10	0·0
11	22·58	0·05	0·054	− 30	− 60	80·0	7·0	1·10	0·5
12	22·46	0·10	0·390	+ 50	+ 80	72·0	4·7	0·80	0·4
13	22·13	0·99	0·054	− 10	− 10	73·7	4·6	0·70	0·2
14	22·52	1·07	0·155	+ 35	− 70	81·0	4·3	0·50	0·6
15	22·72	1·02	0·412	+200	+ 65	0·1	0·0	0·00	0·0
16	22·41	2·64	0·067	+125	+ 40	46·4	1·2	0·60	0·1
17	22·84	2·85	0·159	+170	+150	11·4	2·4	0·50	0·2
18	22·35	2·73	0·310	+920	+130	0·1	0·0	0·00	0·0
19	22·52	2·96	0·435	+940	+905	0·0	0·0	0·00	0·1

Increase in nitrogen content of the molybdenum-free, 17% chromium steel led to some increase in pitting potential (compare results for steels 1 and 2), but improvements from increasing nitrogen were respectively greater for the 17/1 chromium molybdenum (3 to 6) and the 17/3 chromium molybdenum (7 to 10) steels. The same trends were apparent for the 22% chromium steels (molybdenum-free, 11 and 12; 1% molybdenum, 13 to 15; and 3% molybdenum, 16 to 19), but with the 3% molybdenum series there was a very marked improvement in the 'plain' pitting potential at between 0·159% and 0·31% nitrogen, and in the 'crevice' pitting potential at between 0·31% and 0·435% nitrogen. This corresponds to the 'transition' effect noted earlier in alloy development work and described in the introduction. The following summarised values of pitting potential show that a similar transition can be obtained in some circumstances by increase of chromium or molybdenum. The results are for steels with $\sim$0·3% nitrogen.

The trends shown by the potentiodynamic test results can also be seen in less detail in the results for gravimetric testing. In the authors' opinions, such data are best interpreted on the basis

Pitting potential, mV

% Cr* \ % Mo*	0	1	3
17	–25	+50	+95
22	+50	+200	+920

*Nominal values

of whether corrosion has occurred or not, with little emphasis on kinetics which can be influenced by availability of cathodic reactants as well as (if not to the exclusion of) alloy composition. Experience has shown that, with testing of this type, losses of less than 0·2 mg/cm^2 are unlikely to be due to progressive corrosion and thus samples giving such results may be assumed to be resistant to the relevant solutions.

The reasons for the improvement in resistance to pitting corrosion initiation brought about by nitrogen dissolved in the steel can only, at present, be a matter for speculation. One widely held view regarding the passive state, which must be maintained if unprotected ferrous alloys are to be of use in many aqueous environments, is that passivity is due to the presence of a self-healing oxide film which, in the case of the stainless steels, is considered to be chromium-rich. It does not seem possible that an oxide of nitrogen could co-exist with and 'strengthen' such a lattice, but a metal nitride might. The question then is why nitrogen must be dissolved in the steel to have such an effect when there is no shortage of this element in the atmosphere and it is usually present to some degree in aqueous solutions. One possibility is that the increased availability of nitrogen in atomic form at the surface of a steel with a high dissolved nitrogen content could favour nitride formation. The same argument could be applied if an adsorption mechanism of passivity is preferred. Whatever the reason for the observed effects, and it is obvious that much more study is required, it has been found that the use of nitrogen as an alloying element can, under some circumstances, make a substantial practical contribution to the resistance of stainless steels to pitting, which is, perhaps, the most prevalent form of corrosion suffered in practice by these alloys.

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